Ysbyty Gwynedd
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SPECIAL PATIENT GROUPS

Peri-Operative Management of Morbidly Obese Patients

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Key Points

- Increasing prevalence of obesity amongst population (paediatric, obstetric, general)
- Obesity brings challenges, with increased risk of peri-operative morbidity and mortality
- Additional / specialised equipment may be necessary to ameliorate risk to patients/staff
- Drug choices and doses and anaesthetic techniques should be individually planned
- Excellent summary from SOBA on page 3 of this guideline

Introduction

Morbid Obesity is a term previously used to refer to patients with a BMI > 40, or a BMI > 35 with associated co-morbidity. It has been replaced in the WHO classification as below.

BMI	<18.5	18.5 - 25	>25 - 30	>30 - 35	>35 - 40	> 40
Class	Underweight	Normal	Pre-obesity	Obese I	Obese II	Obese III (Morbidly Obese)

Identification of morbidly obese patients pre-operatively

POAC / Elective

- Current height , weight and BMI measurements are recorded in POAC documentation
- Morbid obese patients flagged to list anaesthetist in advance
- Issues raised at team brief at start of each operating list

Obstetrics

BMI 40-45 without co-morbidity

- Elective; assessed on day of admission
- Acute; flagged to 551 bleep holder, +/- duty consultant if requiring caesarean section

BMI > 40 with co-morbidity or > 45

- Antenatal anaesthetic clinic review
- Assessed by duty obstetric anaesthetist on admission on labour suite

Conduct of anaesthesia with morbid obesity

The Society for Obesity and Bariatric Anaesthesia (SOBA) have produced an excellent comprehensive single sheet guideline for the care of the obese patient; this is included on page 3.

Airway / Respiratory

- Ramp position – Ear tragus with with sternal notch. Optimises pre-oxygenation, bag-mask ventilation and intubation.
- Oxford HELP pillows are in obstetric theatre, outside Th11 and between Th5 and Th6
- Difficult airway trolleys and the Fibreoptic Bronchoscope Cabinet are kept between Th5 and Th6.

Positioning

- Following self-positioning on the table the patients should be anaesthetised in theatre.

Drug Doses / Choices

- Increasing obesity gives a disproportionate increase in relatively avascular adipose tissue,
- Use of TBW may result in a relative overdose with bolus/infusions
- Use of LBW may under-dose infusions (i.e. TIVA/Propofol/NMBs)
- Titrate to clinical effect, and use objective end points e.g. quantitative NMB monitors
- Use rapid on/offset anaesthetic agents e.g. desflurane and short acting opioids where possible
- Use local anaesthetics and short acting opioids as part of a multimodal analgesic regime

Total Body Weight (TBW)	• The actual weight of the patient • There will be a disproportionate amount of adipose tissue, which is relatively avascular
Ideal Body Weight (IBW)	• What the patient *would* weigh if they had a healthy ratio of lean to fat mass with maximal life expectancy. A product of age, height, weight and gender. IBW = Height (cm) – 100 (male) or 105 (female)
Lean Body Weight (LBW)	• The patients weight if their fat was excluded • Given by many formulae Generally does not exceed 100kg for males, and 70kg for females
Adjusted Body Weight (ABW)	• ABW takes into account the increased amount of healthy/lean fat obese patients will have over non-obese patients. Calculated by adding 40% of the 'excess' fat to the IBW ABW = (TBW-IBW x 0.4) + IBW

Monitoring / Equipment Considerations

Operating table safe weight limits are available in Theatre 11

Table	Theatres	Maximum Weight (KG)
Schaerer Axis Table	OBS, Th 3-11	180
Maquet Betastar	Th 2	250
Maquet Orthostar II (Traction)	Th 2	225
DRE Anetic-P (DSU Trolley)	DSU	250
Steris Hi Max (Shoulder Table)		Flat 500, Sitting 270

Also: Large BP cuffs, long spinal/epidural needles; vascular access: USS for cannula/CVC insertion, IR Vein finder (Th3); Depth of Anaesthesia Monitoring (BIS); Quantitative Neuromuscular Monitors.

References

- Anaesthesia, Critical Care & Pain Management CPG, Obstetric Anaesthesia Written Control Document, Referral of morbidly obese parturients for anaesthetic assessment (2016)
- AAGBI: Peri-Operative management of the obese surgical patient (2015)

Anaesthesia for the Obese Patient

Society for Obesity and Bariatric Anaesthesia

Pre-operative Evaluation

Red Flags

- Poor functional capacity
- Abnormal ECG
- Uncontrolled BP, CCF or IHD
- SpO2 <94% on air
- If bicarbonate >27, OHS likely
- Previous DVT/PE
- STOP-BANG ≥ 5
- OS-MRS >3
- Metabolic Syndrome
- High NSQIP ACS Risk

Yes →

Consider:

- Preoperative CPAP
- Blood Gases / Sleep Studies
- Echocardiogram
- Cardiorespiratory referral
- Experienced Anaesthetist
- Book HDU Bed

No →

- May be suitable for day-case surgery

OS-MRS Calculator

tools.farmacologiaclinica.info

NSQIP ACS Risk Calculator

riskcalculator.facs.org/RiskCalculator

STOPBANG Calculator

www.stopbang.ca

Central Obesity
(waist > half height)

- Difficult airway/ventilation more likely
- Greater risk of CVS disease/thrombosis
- Higher risk of metabolic syndrome



Peripheral Obesity
(Fat outside body cavity)

- Less co-morbidity
- Lower risk

Intra-operative Management

Suggested Equipment:

- Suitable bed/trolley and operating table
- Gel padding
- Wide strapping
- Table extensions/arm boards
- Forearm cuff or large BP cuff
- Device or equipment for ramping
- Step for anaesthetist
- Difficult airway equipment
- Videolaryngoscope
- Ventilator capable of PEEP & pressure modes
- Hover mattress or equivalent
- Long spinal, regional and vascular needles
- Ultrasound machine
- Appropriately sized calf compression devices
- Depth of anaesthesia monitoring
- Neuromuscular monitoring
- Sufficient staff to move patient

Ramping



- Tragus level with sternum
- Reduces risk of difficult laryngoscopy
- Improves ventilation and pre-oxygenation

Anaesthetic Technique:

- Consider premed antacid & analgesia
- Careful glucose control
- DVT prophylaxis
- Self-position on operating table
- Preoxygenate and intubate in ramped/sitting position
- Consider CPAP and HFNO
- Minimal induction to ventilation time
- Commence maintenance promptly
- Tracheal intubation recommended
- Caution with SAD in BMI >40
- Avoid spontaneous ventilation, use PEEP
- Use short-acting inhalationals or TIVA
- Short-acting opioids & multimodal analgesia
- PONV prophylaxis
- Ensure full NMB reversal
- Extubate and recover sitting up

Lean Body Weight: This exceeds ideal body weight in the obese and plateaus at:

- ≈100kg for a man
- ≈70kg for a woman

Ideal Body Weight: Broca formula

- Men: height (in cm) - 100
- Women: height (in cm) - 105

If in doubt, titrate and monitor effect

Suggested dosing for anaesthetic drugs

Lean Body Weight (Males Max 100Kg Females Max 70Kg)	Adjusted Body Weight (Ideal plus 40% excess)	Total Body Weight
• **Propofol induction** • **Thiopentone** • **Fentanyl and Alfentanil** • **Morphine** • **Non-depolarising NMBDs** • **Paracetamol** • **Local Anaesthetics**	• **Propofol Infusion** • **Neostigmine** (max 5mg) • **Sugammadex** (read pack insert) • **Antibiotics**	**Suxamethonium** **LMWHs** (titrate dose with Xa levels)

Post-operative Care

PACU discharge:

- Usual discharge criteria should be met
- SpO_2 should be maintained at pre-op levels with minimal O_2 therapy
- No evidence of hypoventilation

General good ward level practice includes:

- Multimodal analgesia
- Caution with long-acting opioids and sedatives
- Early mobilisation
- Robust thromboprophylaxis regime
- Experienced Consultant Review

OSA or Obesity Hypoventilation Syndrome:

- Sit up and avoid sedatives and post-op opioids
- Reinstate patient's own CPAP if applicable with additional time in recovery until free of apnoeas without stimulation
- Patients untreated, intolerant of CPAP or ineffectively treated (persistent symptoms) are at risk of hypoventilation
- In these cases, IV opioids should be avoided but where necessary, patient should have continuous SpO_2 monitoring and level 2 care must be considered

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